4-Bit Manchester Carry Look-Ahead Adder Design Using MT-CMOS Domino Logic

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***Abstract*—In this paper, a design of high performance and low power 4-bit Manchester carry look-ahead adder is presented with the help of modified multi-threshold domino logic technique. The introduced MT-MOS transistors decrease the power dissipation of adder circuit by reducing sub threshold leakage current. The FTL dynamic logic is another technique used to increase operating speed of logic circuit, which evaluates the computational blocks partially before the input signals are formalized and then the final evaluation is performed as soon as the input signals arrive. The pre-evaluation logic reduces propagation delay into half. The combination of FTL dynamic logic and MT-CMOS domino logic techniques are yields high fan-out, high switching frequencies since both lower delay and dynamic low power consumption in the arithmetic circuits. A Manchester structure of carry generation employed in carry look-ahead adder minimizes the area of arithmetic circuit by decreasing number of transistors. The simulation results have verified that the proposed techniques are reduced the total power dissipation up to 40%, 55% of propagation delay than the standard dynamic domino CMOS technology.**

***Index Terms*— CLA, CMOS, Domino logic, Dynamic logic, FTL, MT-CMOS**

I. Introduction

The invention of Integrated circuits and MOS transistors are brought a giant technological leap in microelectronics. The increased performance, and reduced power consumption and area are importantly notable in integrated circuits. Also the increasing demands of high performance processors are developing complicated rise in technological emergent of integrated circuits. This emerging growth is forcing the researchers to develop the new dynamic circuits to improve the performance, area efficiency, and functionality of different logical and arithmetic circuits. Normally, in the integrated circuits arithmetic operations are widely used for basic and enhanced applications. Many techniques were developed to improve the performance of arithmetic circuit designs. So far the use of carry look-ahead principle of high-speed arithmetic unit is precisely one of the challenges in VLSI processor design structured today for multilevel CLA circuits without limiting the functional flexibility.

The domino CMOS logic is a popularly used design technique in high performance processors which offer smaller area and higher performance than complementary CMOS design. However the average power consumption of domino logic is greater than static circuit. A low-power dual threshold voltage-voltage scaling technique for domino logic circuit [1] reduces the total power consumption. Also the multi threshold voltage transistors prevent the circuit from unwanted switching transients. In [2], a dual threshold voltage technique for glitch minimization is presented and verified with CMOS logic circuits. The high threshold voltage transistors are not sensitive to the leakage current. Also the total power consumption of the circuit can be reduced through high threshold voltage transistors considerably. It is verified in [3] with the name of NBTI-aware dual threshold voltage assignment for leakage power reduction. In paper [4], an FTL (Feed through logic) design technique is proposed for high performance dynamic CMOS logic with high noise immunity and low power. An FTL improves the performance of arithmetic circuits with a very long logic depth. The dual V_T techniques are proposed in [5] and the outputs are verified as low power circuits by comparing with conventional CMOS digital circuits. Chen and Li [6] have developed an improved 32-bit carry look-ahead adder with Conditional Carry-Selection for high speed and low power arithmetic circuit. It is implemented on silicon to verify the power reduction as well as the preservation of high speed. In [7] Levi, Bass have developed a high speed dual mode logic carry look ahead adder to limit the power dissipation of logic circuits. The output of Multi-threshold CMOS design for low power digital circuit is proposed and verified in [8]. Sleep switch dual threshold voltage domino logic with reduced sub threshold and gate oxide leakage current is proposed and verified in [9] by Wang, Huang, and Cheng. The reduction of sub threshold leakage and gate oxide leakage current are achieved in [10] for domino logic circuit.

Domino logic technique improves the performance and reduces the device count and silicon area of arithmetic circuit when compared to the standard fully complementary static CMOS logic. However, the major drawback with the domino dynamic logic circuit is that the circuit consumes more power due to the switching activity and the clock load of the circuit. To eliminate the excessive power dissipation of the dynamic logic, the current design methodologies trade the power for performance in critical delay sections of arithmetic circuit. This technique can be achieved through a combination of dynamic and static circuit styles, and by the use of dual supply voltages and dual-V_T transistors. So far domino and dynamic logics are endured by charge sharing and race problems. To solve these demerits, we are proposing the new MT-CMOS dynamic logic circuit using multi threshold MOS transistor for reducing the power dissipation. Also to increase the performance of the dynamic logic circuit, FTL logic technique is employed in the arithmetic circuits. The proposed circuit is giving solution for problems discussed and also minimizing power wastages. In this paper, we are also proposing a low power high performance 4-bit Manchester carry look-ahead logic structure

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using a new MT-CMOS domino logic blocks to reduce device count.

II. Principles of Operation

A. MT-CMOS Domino Logic

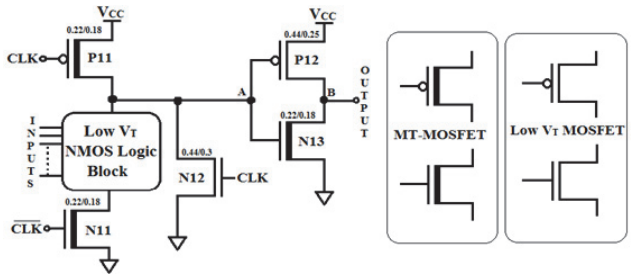


Fig. 1. Proposed MT-CMOS domino logic

The proposed MT-CMOS Domino Logic Circuit is shown in Fig 1. In the circuit, multi threshold MOS transistors are printed in dark line on gate to differentiate from low threshold voltage transistors. The employment of MT-CMOS transistors in domino logic circuits reduce the power dissipation by eliminating sub threshold leakage current and gate oxide leakage current with high threshold voltage. Normally multi threshold transistors i.e. MT-PMOS and MT-NMOS transistors are larger than standard transistors. This condition greatly depends on the parasitic capacitors which present in the transistor. When the PMOS and NMOS transistor width is reduced, the parasitic capacitances become larger and that tends to suppress the switching time of logic low and high to improve the speed. In the standby (Sleep) mode, i.e., when logic high clock signal is applied to circuit, the MT-transistors switch to off, that leads the circuit for reduction of static power dissipation and elimination of sub threshold current. It is reducing the total power dissipation of MT-CMOS circuit. In evaluation mode, dynamic logic circuit improves the speed of switching operation through feed through logic.

When the CLK is low, the transistors P11 and N11 are ON. This is known as evaluation phase. During this evaluation phase, the low V_T NMOS logic block is evaluated efficiently to switch the node voltage V_A almost equal to VCC or ground. The low V_T logic circuit has reduced logic delay by 55% compared to high-V_T logic circuit. Hence it is evaluated faster to produce the output. The proposed logic circuit provides the flexibility to design the logic circuit, so it can be used to design multistage domino logic with iterative circuits.

The two stage MT-CMOS dynamic logic circuit shown in Fig 2 is expressing the cascaded connection of proposed technique to construct pipeline structure. In this two stage implementation, stage 1 and 2 are operating alternatively in pre-charging phase and evaluation phase. If the applied CLK is high, first stage operates in pre-charging phase and second stage operates in evaluation phase. MT-NMOS N22 is used to hold the output of first stage in logic high during the pre-charging phase. This logic high output is applied to evaluate low V_T NMOS section of second stage. The second stage evaluates the low V_T NMOS logic block by activating MT-transistors P23 and N24. If the applied CLK is low, first stage operates in evaluation phase and second stage operates in pre-charging phase. The first stage evaluates the low V_T NMOS logic block by the activation of the MT-Transistors P21 and N21. At the same time the second stage operates in pre-charging phase. The signal applied into MT-NMOS N25 is used to hold the output of second stage in logic high during the pre-charging phase which evaluates the next stage. This alternate switching of pre-charging and evaluation of adjacent stages is also assisting to reduce the power consumption of circuit into one-half. Also this FTL based dynamic domino logic operates faster to develop the output when compare to typical CMOS dynamic logic circuit.

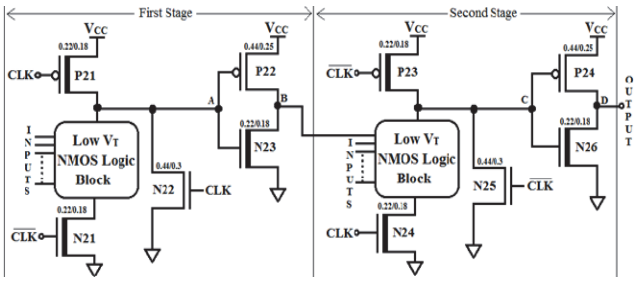


Fig. 2. Cascaded two stage MT-CMOS dynamic logic

B. Manchester Carry Look-Ahead Adder

Carry look-ahead adder works with a principle of carry generation and propagation. It allows the preprocessing addition step of two numbers to find the carry. That helps to reduce the propagation delay of adder. Normally the efficiency of domino carry chain is enriched by pre-charging phase of circuit in appropriate points. The adder circuit works in pre-charging phase when the clock input is low. When MT-transistors are turned off, the output goes to logic high through inverter. When the clock input is high, the adder circuit operates in evaluation phase. When MT-transistors are turned on, the logic blocks are evaluated by the inputs applied to low V_T logic block. However, the carry look-ahead adder uses many transistors to construct a logic block. If numbers of bits are increased in the adder, the transistors count also increases maximally in an adder circuit. Here, we are introducing a new Manchester carry look-ahead structure to reduce the number of elements employed in the construction of adder circuit. Manchester carry chain is a carry look-ahead generator that uses shared logic technique to reduce the transistor count.

The circuit of 4-bit Manchester carry look-ahead adder is shown in Fig 3. Fig 3(a) and 3(b) are representing the circuit of propagate and generate signal provider of the adder. The logic blocks are implemented by new FTL MT-MOS transistors to reduce the propagation delay and power dissipation. Fig 3(c) is a circuit of sum calculator which is also constructed by proposed MT-MOS transistor technique. It develops the sum output as soon as the carry and propagate signals arrive from the carry and propagate signal providers. Fig 3(d) shows the circuits of Manchester carry chain which generates various carry signals. The carry signal of adder is obtained through an inverter. However no longer we need the intermediate carry signals as the carry bits distributed in the circuit arrangement. A fourth bit of carry propagation of Manchester structure is

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chosen to reduce numbers of series propagate transistors which reduces the consequence of body effect.

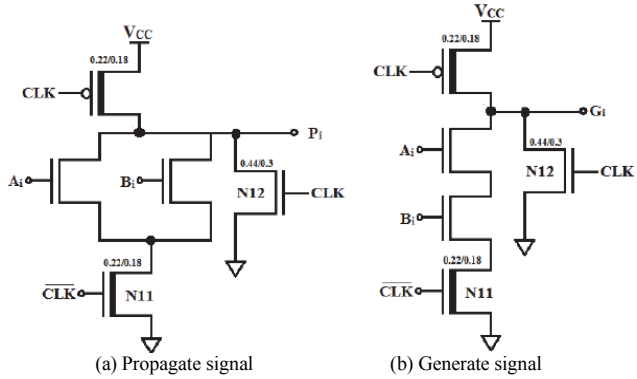
The logic expressions of proposed carry look-ahead adder define the logic output of adder circuit. The logic structure of a proposed Manchester carry look-ahead adder is evaluated to determine the logic expressions. The logic sum and carry output of the above circuits are obtained as follows,

$$C_i = G_i + P_i.C_{i-1}$$
$$S_i = P_i \oplus \overline{C_{i-1}} \quad (1)$$

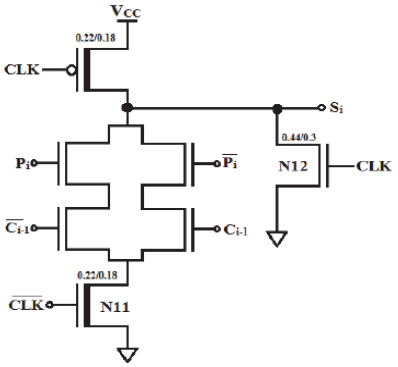
Where,

$$G_i = A_i.B_i \quad (2)$$

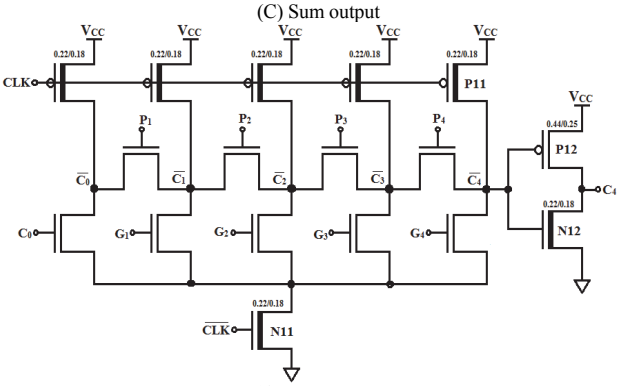
$$P_i = A_i + B_i \quad (3)$$



(a) Propagate signal (b) Generate signal



(C) Sum output

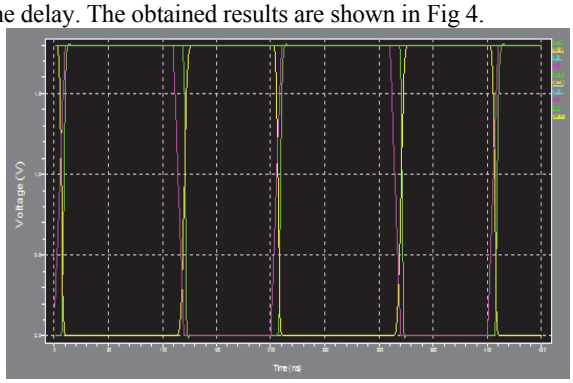


(d) Carry output

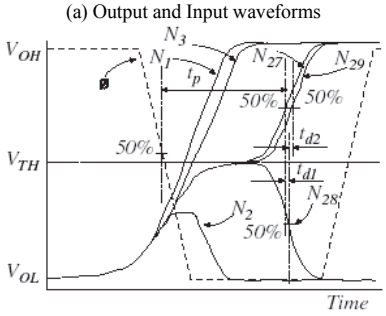
Fig. 3. Circuit of Manchestor Carry Look-ahead adder

III. Simulation Results

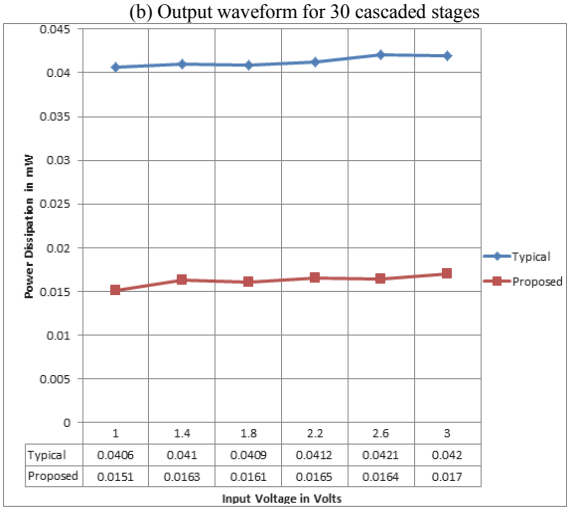
The proposed high performance and low power MT-CMOS based 4-bit Manchester carry look-ahead adder technique is increasing the speed and reducing the average power consumption of the circuit. To verify this assertion, the simulations were conducted for the proposed MT-CMOS dynamic logic and typical CMOS dynamic logic circuits. A TSMC 0.18µm CMOS technology was employed in tanner EDA tool to determine the power dissipation and operating delay of circuits. The proposed domino logic circuit is simulated and verified for power reduction that ensures the reduced power dissipation compared to the typical domino logic circuit. Also the proposed Manchester 4-bit carry look-ahead adder method is simulated and verified with tanner tool which results the reduced power dissipation and propagation time delay. The obtained results are shown in Fig 4.



(a) Output and Input waveforms



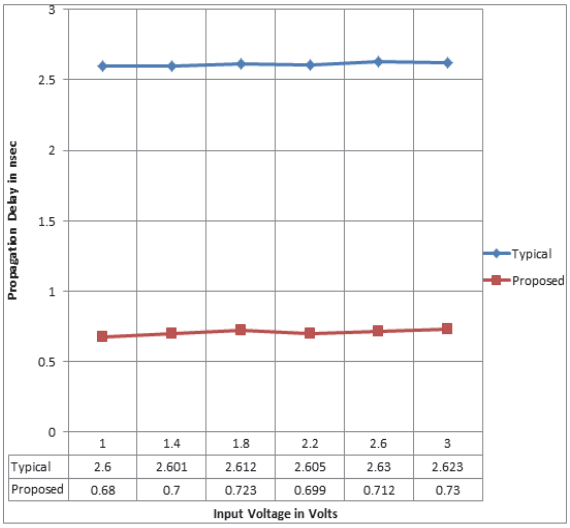
(b) Output waveform for 30 cascaded stages



	1	1.4	1.8	2.2	2.6	3
Typical	0.0406	0.041	0.0409	0.0412	0.0421	0.042
Proposed	0.0151	0.0163	0.0161	0.0165	0.0164	0.017

(c) Power Analysis Chart

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	1	1.4	1.8	2.2	2.6	3
Typical	2.6	2.601	2.612	2.605	2.63	2.623
Proposed	0.68	0.7	0.723	0.699	0.712	0.73

(d) Delay Analysis Chart

Fig. 4. Simulation results

The power and delay values are also verified using following formulas. Transient power consumption can be calculated using the equation:

$$\text{Power}(P_T) = C_{pd} \times V_{CC}^2 \times F_I \times N_{SW} \quad (4)$$

Where,

P_T → Transient power consumption;
V_{CC} → Supply voltage;
F_I → Input signal frequency;
N_{SW} →Number of bits switching; and
C_{pd} → Dynamic power-dissipation capacitance.

Figure of merit =delay × power × area.

Average propagation delay (t_{pd}),

$$t_{pd} = \frac{(t_{pdr} + t_{pdf})}{2} \quad (5)$$

Where,

t_{pdr}→ rising propagation delay, From input to rising output crossing VDD/2.

t_{pdf} → falling propagation delay, From input to falling output crossing VDD/2.

The input/output waveforms and power and area comparison chart are shown in Fig 4.

IV. Conclusion

In this paper, we have presented an efficient method with minimized power and increased speed of dynamic logic circuit for arithmetic circuits using FTL and MT-CMOS logic techniques. Charge sharing problems associated with dynamic and domino families are also removed in the proposed structures. By eliminating these power wastages, the total power consumption of adder circuits is reduced with the help of MT-CMOS domino logic technique. The propagation delay of adder is also decreased by pre-evaluation procedure of FTL which helps to increase the operating speed of adder. Experiments have ensured the high speed operation and low power dissipation of Manchester carry look-ahead adder by the proposed model match with SPICE simulation results. For the experimental verification, the proposed Manchester 4-bit carry look-ahead adder method is simulated in tanner EDA tool with TSMC 0.18µm technology file. Hence, the simulated result shows that the reduction of power dissipation up to 40% and propagation time delay up to 55%.

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